

TrES-3b

R-1729

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_210430 · created 2026-08-02T17:25:06+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459334.8988 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.166 +/- 0.037
Transit depth (Rp/Rs)^2	2.76 +/- 1.23 [%]
Orbital Inclination (inc)	81.27 +/- 1.41
Airmass coefficient 1 (a1)	1.088 +/- 0.012
Airmass coefficient 2 (a2)	-0.078 +/- 0.023
Scatter in the residuals of the lightcurve fit is	1.06 %
Best Comparison Star	#2 - [315, 59]
Optimal Method	PSF photometry
Transit Duration (day)	0.045 +/- 0.02

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.166 ± 0.037 vs 0.16309 (deviation 0.08σ)
Duration fit vs published	0.045 d vs 0.0567 d (deviation 0.58σ)
Mid-time O–C	1.1 min
Depth SNR	4.5
Residual scatter	1.06 %

Light curve

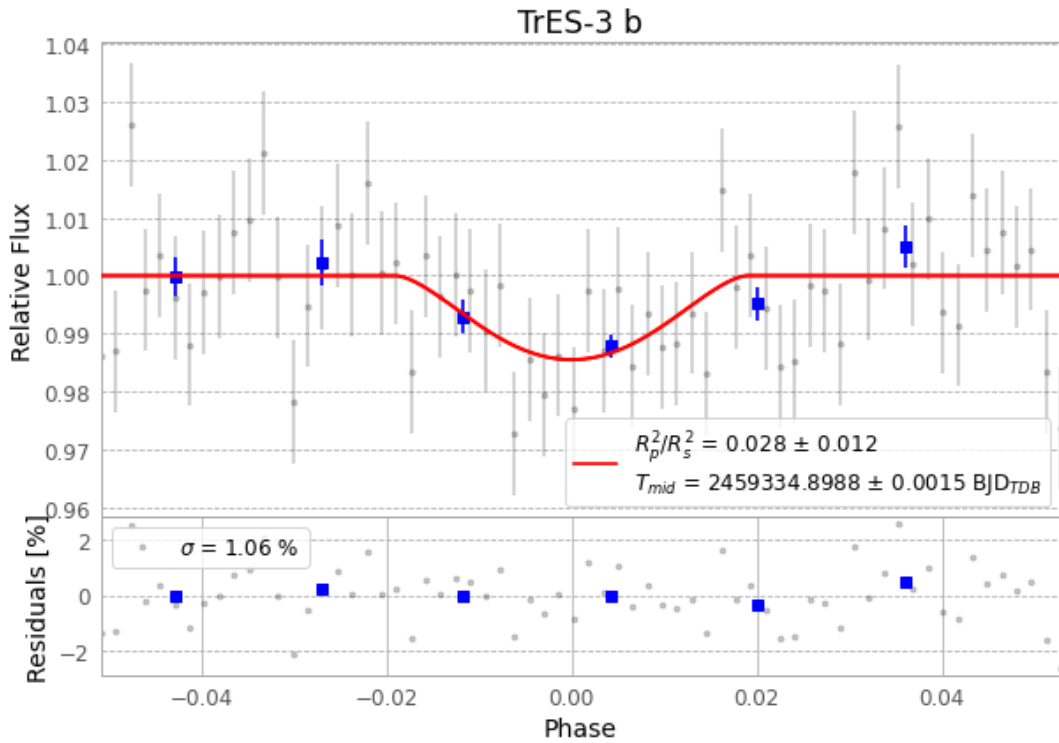


Figure 1 — FinalLightCurve_TrES-3 b_2021-04-30.png (SHA-256 ebcd173d7573ecc...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [384, 220], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 483b4fe6fe413f18...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

66 FITS frames (43 MB), TRES-3210430075412.FITS → TRES-3210430110912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-210430.log (c62a2a11adb3...), FinalLightCurve_TrES-3 b_2021-04-30.png (ebcd173d7573...), FinalLightCurve_TrES-3 b_2021-04-30.csv (f2808c25d9d9...)

Compute resources

Wall time 150.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 17:25Z.